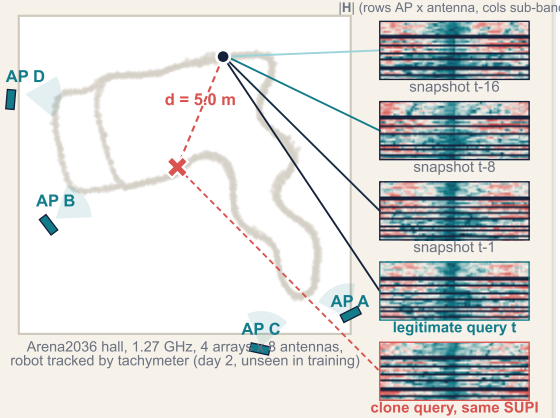
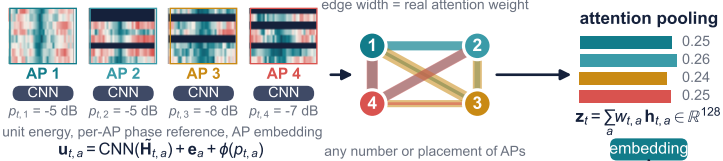
A a measured cell-free cluster

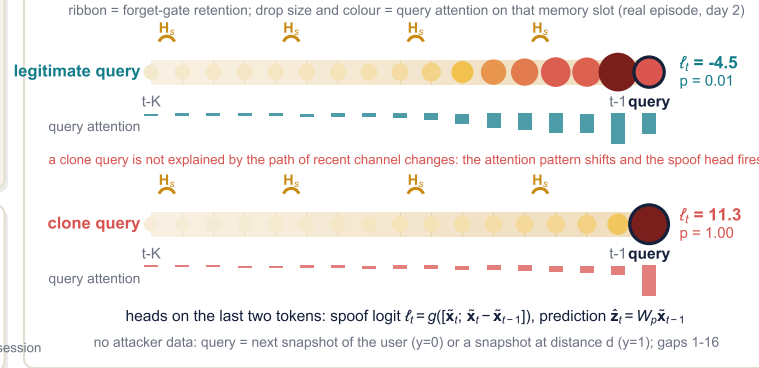


Arena2036 hall, 1.27 GHz, 4 arrays, 8 antennas,
robot tracked by tachymeter (day 2, unseen in training)

B per-snapshot encoder: AP tokens, set attention, pooling

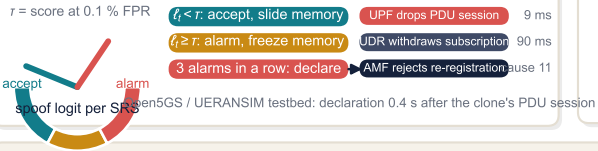


C channel memory: path-encoded forgetting attention



no attacker data: query = next snapshot of the user (y=0) or a snapshot at distance d (y=1); gaps 1-16

D decision and core-network response



E the memory step, in one line

$$a_{t,s} = \frac{\mathbf{q}_t^\top \mathbf{H}_t \mathbf{H}_{t-1} \cdots \mathbf{H}_{s+1}\,\mathbf{k}_s}{\sqrt{d_h}} + \sum_{r=s+1}^{t} \log f_r, \qquad \mathbf{H}_s = \mathbf{I} - \beta_s \mathbf{w}_s \mathbf{w}_s^\top,\ \ \beta_s = 2\sigma(\mathbf{a}^\top \mathbf{x}_s),\ \ \|\mathbf{w}_s\| = 1, \qquad f_r = \sigma(\mathbf{w}_f^\top \mathbf{x}_r)$$

$$\alpha_{t,s} = \mathrm{softmax}_{s \le t}(a_{t,s}), \qquad \mathbf{x}_t \leftarrow \mathbf{x}_t + W_o \sum_{s \le t} \alpha_{t,s}\,\mathbf{v}_s, \qquad L = \mathrm{BCE}(\ell_t, y) + \lambda(1 - \cos(\hat{\mathbf{z}}_t, \mathbf{z}_t))\,\mathbb{1}[y = 0]$$

the key of an old snapshot is carried to the query through the Householder transforms of every snapshot between them,
so the logit follows the path of channel changes rather than elapsed time; the forget term discounts snapshots behind abrupt changes

F evaluation: three datasets, two reference tests, eight ablations, one protocol-level testbed

DICHASUS cell-free (measured)	Munich 28 GHz (Sionna RT)	San Francisco 7 GHz (Sionna RT)	reference tests (training-free)	ablations (same episodes, seed)
1.27 GHz, 4 x 8 ant., 2 days, 68k snapshots	6 APs x 8 ant., 240 trajectories, FR2	6 APs x 8 ant., 240 trajectories, FR3	AP-weighted correlation, AP power profile	RoPE / no positions, no forget gate, mean AP pooling, K = 4/8/32, no prediction loss

clone distance 0.25 m - 400 m, trajectory-disjoint splits; day 2 of the measurements is used only for testing and for the testbed replay